

- 1. Prerequisites** A knowledge of the specifications for P1 and P2, their prerequisites and associated formulae, is assumed and may be tested.
- 2. Examination**
- First assessment: January 2020.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 75 marks.
 - Students must answer all questions.
 - Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
 - The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
- 3. Notation and formulae** Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.
- Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Trigonometry

$$\cos^2 A + \sin^2 A \equiv 1$$

$$\sec^2 A \equiv 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A \equiv 1 + \cot^2 A$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

3. Notation and formulae Differentiation

continued

$f(x)$	$f'(x)$
$\sin kx$	$k \cos kx$
$\cos kx$	$-k \sin kx$
e^{kx}	ke^{kx}
$\ln x$	$\frac{1}{x}$
$f(x) + g(x)$	$f'(x) + g'(x)$
$f(x) g(x)$	$f'(x) g(x) + f(x) g'(x)$
$f(g(x))$	$f'(g(x)) g'(x)$
a^x	$a^x \ln a$

Integration

$f(x)$	$\int f(x) \, dx$
$\cos kx$	$\frac{1}{k} \sin kx + c$
$\sin kx$	$-\frac{1}{k} \cos kx + c$
e^{kx}	$\frac{1}{k} e^{kx} + c$
$\frac{1}{x}$	$\ln x + c, x \neq 0$
$f'(x) + g'(x)$	$f(x) + g(x) + c$
$f'(g(x)) g'(x)$	$f(g(x)) + c$
a^x	$\frac{a^x}{\ln a} + c$

P3.3 Unit content

What students need to learn:		Guidance
1. Algebra and functions		
1.1	Simplification of rational expressions including factorising and cancelling, and algebraic division.	Denominators of rational expressions will be linear or quadratic, e.g. $\frac{1}{ax+b}$, $\frac{ax+b}{px^2+qx+r}$, $\frac{x^3+1}{x^2-1}$.
1.2	Definition of a function. Domain and range of functions. Composition of functions. Inverse functions and their graphs.	The concept of a function as a one-one or many-one mapping from $\mathbb{R}$ (or a subset of $\mathbb{R}$) to $\mathbb{R}$. The notation $f: x \mapsto$ and $f(x)$ will be used. Students should know that fg will mean ‘do g first, then f ’. Students should know that if f^{-1} exists, then $f^{-1}f(x) = ff^{-1}(x) = x$.
1.3	The modulus function.	Students should be able to sketch the graphs of $y = ax + b $ and the graphs of $y = f(x) $ and $y = f(x)$, given the graph of $y = f(x)$. For example, sketch the graph with equation $y = 2x - 1 $ and use the graph to solve the equation $ 2x - 1 = x + 5$ or the inequality $ 2x - 1 > x + 5$.
1.4	Combinations of the transformations $y = f(x)$ as represented by $y = af(x)$, $y = f(x) + a$, $y = f(x + a)$, $y = f(ax)$.	Students should be able to sketch the graph of, for example, $y = 2f(3x)$, $y = f(-x) + 1$, given the graph of $y = f(x)$ or the graph of, for example, $y = 3 + \sin 2x$, $y = -\cos\left(x + \frac{\pi}{4}\right)$. The graph of $y = f(ax + b)$ will <i>not</i> be required.